Understanding Analysis, 2nd Edition
Chapter 5 - Section 5.3 - The Mean Value Theorems

1. Recall from Exercise 4.4.9 that a function $f : A \to \mathbf{R}$ is Lipschitz on A if there exists an $M > 0$ such that
$$\left|\frac{f(x) - f(y)}{x - y}\right| \leq M$$
for all $x \neq y$ in A.

   (a) Show that if f is differentiable on a closed interval $[a, b]$ and if f' is continuous on $[a, b]$, then f is Lipschitz on $[a, b]$.

   (b) Review the definition of a contractive function in Exercise 4.3.11. If we add the assumption that $|f'(x)| < 1$ on $[a, b]$, does it follow that f is contractive on this set?

Solution:

(a) Continuous function preserves compactness (Theorem 4.4.1) and compact sets are bounded. Since f' is continuous on $[a, b]$, it's bounded. Thus there exists $M > 0$ such that $|f'(c)| \leq M$ for all $c \in [a, b]$.

Now, by the Mean Value Theorem, for any $x \neq y$ in $[a, b]$, there exists t in $[a, b]$ such that
$$\frac{f(x) - f(y)}{x - y} = f'(t)$$
Therefore
$$\left|\frac{f(x) - f(y)}{x - y}\right| = |f'(t)| \leq M$$
as required.

(b) f is contractive if there exists a constant $0 \leq c < 1$ such that $|f(x) - f(y)| \leq c|x - y|$ for all $x, y \in \mathbf{A}$.

The answer is yes. From part (a), we have
$$|f(x) - f(y)| = |f'(t)||x - y|$$
And we have $|f'(t)| < 1$.

But we need to choose a single constant c (whereas $|f'(t)|$ depends on x and y). From part (a), f' is continuous on a compact set $[a, b]$ so it attains a maximum.

Choose $c = \max_{x\in[a,b]} |f'(x)|$. Since $|f'(x)| < 1$ on $[a,b]$ from the problem's assumption, the contractive condition is satisfied.

2. Let f be differentiable on an interval A. If $f'(x) \neq 0$ on A, show that f is one-to-one on A. Provide an example to show that the converse statement need not be true.

Solution: We prove the contrapositive. Suppose f is not one-to-one, then there exists $a < b$ in A such that $f(a) = f(b)$. By the Mean Value Theorem, there exists $c \in A$ such that

$$\frac{f(b) - f(a)}{b - a} = 0 = f'(c)$$

so f' cannot be non-zero on all of A.

Counterexample to the converse: x^2 on $[0,1]$. It's one-to-one (because we exclude negative domain) with $f'(0) = 0$.

Alternatively, a cleaner counterexample that doesn't involve one-sided derivative is x^3.

3. Let h be a differentiable function defined on the interval $[0,3]$, and assume that $h(0) = 1$, $h(1) = 2$, and $h(3) = 2$.

   (a) Argue that there exists a point $d \in [0,3]$ where $h(d) = d$.

   (b) Argue that at some point c we have $h'(c) = 1/3$.

   (c) Argue that $h'(x) = 1/4$ at some point in the domain.

Solution:

(a) The fixed point d is not necessarily in $[0,1]$, as we can draw a straight line between $h(0)$ and $h(1)$ and there'd be no fixed point. So it must be in $[1,3]$.

The proof is basically the same as Exercise 4.5.7. Define $g(x) = h(x) - x$ on $[1,3]$. If there exists a fixed point d then it must be that $g(d) = 0$. We have $g(1) = 1$ and $g(3) = -1$. g is continuous, so we can use IVT to conclude that there exists such a d.

(b) The slope between $x = 0$ and $x = 3$ is $\frac{h(3)-h(0)}{3-0} = 1/3$. By the Mean Value Theorem, there exists a $c \in (0,3)$ such that $h'(c) = 1/3$.

(c) Applying the same argument as part (b) to $x = 1$ and $x = 3$, we can conclude that there exists $d \in (1, 3)$ such that $h'(d) = 0$. Now we have

$$(h'(d) = 0) < 1/4 < (h'(c) = 1/3)$$

Apply Darboux Theorem to conclude that there exists $x \in (0, 3)$ such that $h'(x) = 1/4$.

4. Let f be differentiable on an interval A containing zero, and assume (x_n) is a sequence in A with $(x_n) \to 0$ and $x_n \neq 0$.

   (a) If $f(x_n) = 0$ for all $n \in \mathbf{N}$, show $f(0) = 0$ and $f'(0) = 0$.

   (b) Add the assumption that f is twice-differentiable at zero and show that $f''(0) = 0$ as well.

Solution:

(a) $f(x_n)$ is the zero sequence, so we have $(x_n) \to 0 \implies f(x_n) \to 0$. Since f is differentiable on A, it's continuous on A, so we have $f(x_n) \to f(0) = 0$.

Note that it's enough to prove this for the one sequence (x_n) instead of *any* sequence that converges to zero, because we're given that f is differentiable, so its limit is well defined and any sequence that converges to zero has the same functional limit.

For $f'(0)$, we compute it from the definition. Using the fact that $f(x_n) = 0$

$$f'(0) = \lim_{n\to\infty} \frac{f(x_n) - f(0)}{x_n - 0} = \lim_{n\to\infty} 0/x_n = 0$$

Again, it's enough to prove this for a single sequence (x_n) because we know in advance that f is differntiable. Any sequence that converges to zero must produce the same derivative.

(b) The second derivative, combined with the fact that $f'(0) = 0$, is

$$f''(0) = \lim_{x\to 0} \frac{f'(x) - f'(0)}{x} = \lim_{x\to 0} \frac{f'(x)}{x}$$

To prove this is zero, we need to construct a sequence $(c_n) \to 0$ such that $f'(c_n)$ is eventually all zero. Just like part (a), constructing one such sequence is enough because we're given f is twice-differentiable.

We can construct (c_n) out of (x_n). First, choose a subsequence (y_n) out of (x_n) such that $y_n \neq y_{n+1}$ for all n, and also that all (y_n) lie on one side of zero.

Now, $f(y_n) = 0$ for all n, the slope between $f(y_n)$ and $f(y_{n+1})$ is zero. By MVT (which we can apply because $y_n \neq y_{n+1}$), there exists $c_n \in (y_n, y_{n+1})$ such that $f'(c_n) = 0$. Also, $c_n \neq 0$ (because y_n and y_{n+1} are on the same side of zero).

Now we have $(c_n) \to 0$, $c_n \neq 0$ and $f'(c_n) = 0$ for all $n \in \mathbf{N}$. The second derivative at zero is

$$f''(0) = \lim_{n\to\infty} \frac{f'(c_n)}{c_n} = \lim_{n\to\infty} \frac{0}{c_n} = 0$$

5. (a) Supply the details for the proof of Cauchy's Generalized Mean Value Theorem (Theorem 5.3.5).

   (b) Give a graphical interpretation of the Generalized Mean Value Theorem analogous to the one given for the Mean Value Theorem at the beginning of Section 5.3. (Consider f and g as parametric equations for a curve).

Solution:

(a) Define

$$h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$$

We have $h(a) = h(b)$, so we can apply Rolle's Theorem to conclude that there exists $c \in (a, b)$ such that

$$h'(c) = 0 = [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c)]$$

And rearranging gives $[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c)$ as required. If $g'(c) \neq 0$, we can also write it as $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

(b) Consider the curve defined by the parametric equation $(g(t), f(t))$. Note the ordering, g for the x-axis and f for the y-axis.

A line from $(g(a), f(a))$ to $(g(b), f(b))$ has the slope $\frac{f(b)-f(a)}{g(b)-g(a)}$. This is the rhs of the Theorem (in its fractional form).

The derivative of the curve at a point c is given by

$$\lim_{t \to c} \frac{f(t) - f(c)}{g(t) - g(c)} = \lim_{t \to c} \frac{[f(t) - f(c)] * \frac{1}{t-c}}{[g(t) - g(c)] * \frac{1}{t-c}}$$
$$= \frac{f'(c)}{g'(c)}$$

and this is the lhs of the Theorem.

In conclusion, given a line from $(g(a), f(a))$ to $(g(b), f(b))$, the Generalized Mean Value Theorem says that there exists a tangent to the curve at $(g(c), f(c))$ with the same slope (i.e. parallel) to the line, provided $g'(c) \neq 0$.

The regular Mean Value Theorem is a special case where $g(t) = t$.

6. (a) Let $g : [0, a] \to \mathbf{R}$ be differentiable, $g(0) = 0$, and $|g'(x)| \le M$ for all $x \in [0, a]$. Show $|g(x)| \le Mx$ for all $x \in [0, a]$.

   (b) Let $h : [0, a] \to \mathbf{R}$ be twice differentiable, $h'(0) = h(0) = 0$ and $|h''(x)| \le M$ for all $x \in [0, a]$. Show $|h(x)| \le Mx^2/2$ for all $x \in [0, a]$.

   (c) Conjecture and prove an analogous result for a function that is differentiable three times on $[0, a]$.

Solution:

(a) For $g(0)$ we have $|g(0)| = 0 = M \cdot 0$ as required.

For $x \neq 0$, applying the Mean Value Theorem with the point 0 and $0 < x \le a$, then there exists $c \in (0, x)$ such that

$$\frac{g(x) - g(0)}{x - 0} = \frac{g(x)}{x} = g'(c)$$

Since $|g'(x)| \le M$ for all $x \in [0, a]$, we have

$$|g'(c)| = \left|\frac{g(x)}{x}\right| \le M \implies |g(x)| \le Mx$$

as required.

(b) **(TODO unfinished)** Like part (a), apply MVT to the second derivative

$$\frac{h'(x) - h(0)}{x - 0} = \frac{h'(x)}{x} = h''(c)$$

Take absolute value

$$|h''(c)| = \left|\frac{h'(x)}{x}\right| \le M \implies |h'(x)| \le Mx$$

7. A fixed point of a function f is a value x where $f(x) = x$. Show that if f is differentiable on an interval with $f'(x) \neq 1$, then f can have at most one fixed point.

Solution: We prove the contrapositive. Suppose that f has two fixed points x and y. By MVT, there exists c such that

$$f'(c) = \frac{f(x) - f(y)}{x - y} = \frac{x - y}{x - y} = 1$$

8. Assume f is continuous on an interval containing zero and differentiable for all $x \neq 0$. If $\lim_{x\to 0} f'(x) = L$, show $f'(0)$ exists and equals L.

Solution: Expanding $\lim_{x\to 0} f'(x) = L$ in $\epsilon - \delta$: given $\epsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x| < \delta \implies |f'(x) - L| < \epsilon$$

The same δ works to prove that $f'(0) = L$, because if $|c| < \delta$ then by MVT there exists $|x|$ between 0 and $|c|$ such that

$$0 < |x| < |c| < \delta \implies \frac{f(c) - f(0)}{c} = f'(x)$$

so

$$|f'(x) - L| < \epsilon \implies \left| \frac{f(c) - f(0)}{c} - L \right| < \epsilon$$

— **Doesn't work** —

This was my first approach, but I found out in the end that it doesn't work. The idea is the same as proving $\lim_{b\to c}[\lim_{a\to b} f(a)] = \lim_{a\to c} f(a)$ (given all the limits are defined). Make b close to c and a close to b, then a is close to c.

We're given

$$\lim_{x\to 0} f'(x) = L$$

Choose x small enough such that derivatives near x is close to L

$$|x| < \delta \implies |f'(x) - L| < \epsilon$$

Expanding the definition of $f'(x)$

$$f'(x) = \lim_{t\to x} \frac{f(t) - f(x)}{t - x}$$

The derivative is defined for all $x \neq 0$. Choose t close enough to x such that the sequence of slopes is close to $f'(x)$

$$|t - x| < \delta \implies \left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| < \epsilon$$

Combining both choices of δ, we have the distance between t and 0 is

$$|t - 0| \leq |t - x + x - 0| \leq |t - x| + |x| < 2\delta$$

and the distance between the slope with L is

$$\begin{aligned}\left|\frac{f(t)-f(x)}{t-x}-L\right| &\leq \left|\frac{f(t)-f(x)}{t-x}-f'(x)+f'(x)-L\right| \\ &\leq \left|\frac{f(t)-f(x)}{t-x}-f'(x)\right|+|f'(x)-L| \\ &< 2\epsilon\end{aligned}$$

...aandd it's useless. What we need is the slope between $f(t)$ and $f(0)$, not $f(t)$ and $f(x)$. Instead, we'll need MVT to relate the $f(t)-f(0)$ slope with derivatives at non-zero points.

9. Assume f and g are as described in Theorem 5.3.6 but add the assumption that f and g are differentiable at a, and f' and g' are continuous at a with $g'(a) \neq 0$. Find a short proof for the 0/0 case of L'Hospital's Rule under this stronger hypothesis.

Solution: To restate the Rule with the added hypothesis: let f and g be differentiable on an interval containing a with f' and g' continuous at a. If $f(a) = g(a) = 0$ and $g'(x) \neq 0$ on the interval, then

$$\lim_{x\to a}\frac{f'(x)}{g'(x)} = L \implies \lim_{x\to a}\frac{f(x)}{g(x)} = L$$

Now the proof. Since f' and g' are continuous at a, we have

$$\lim_{x\to a}\frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)} = L$$

$g'(a) \neq 0$ so the fraction is defined. Expanding $f'(a)$ and $g'(a)$ with the definition of derivative

$$\lim_{x\to a}\frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}} = \lim_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)} = L$$

And finally, using the given assumption that $f(a) = g(a) = 0$ completes the proof.

10. Let $f(x) = x\sin(1/x^4)e^{-1/x^2}$ and $g(x) = e^{1/x^2}$. Using the familiar properties of these functions, compute the limit as x approaches zero of $f(x), g(x), f(x)/g(x)$ and $f'(x)/g'(x)$. Explain why the results are surprising but no in conflict with the content of Theorem 5.3.6.

Solution: TODO not finished. As x approaches zero:

- The limit of $\sin(1/x^4)$ is undefined but is bounded by $[-1, 1]$, so $x\sin(1/x^4) \leq |x|$ and by squeeze converges to 0.
- $e^{1/x^2} \to \infty$ because $e > 1$, therefore $e^{-1/x^2} \to 0$.

All in all, $f(x) \to 0$ and $g(x) \to \infty$.

11. (a) Use the Generalized Mean Value Theorem to furnish a proof of the 0/0 case of the L'Hospital's Rule (Theorem 5.3.6).

    (b) If we keep the first part of the hypothesis of Theorem 5.3.6 the same but we assume that
    $$\lim_{x\to a} \frac{f'(x)}{g'(x)} = \infty,$$
    does it necessarily follow that
    $$\lim_{x\to a} \frac{f(x)}{g(x)} = \infty?$$

Solution:

(a) For every $x > a$ (the case $x < a$ is similar), by GMVT there exists $x < c_x < a$ such that
$$\frac{f'(c_x)}{g'(c_x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f(x)}{g(x)}$$
where we used the fact that $f(a) = g(a) = 0$. The denominators are all non-zero. $g'(c_x) \neq 0$ by the Theorem's assumption, and $g(x) - g(a)$ cannot be zero because otherwise Rolle's Theorem gives a $g'(c_x) = 0$, contradicting the Theorem's assumption.

Now if we take $x \to a$, then $c_x \to a$ by Squeeze Theorem. Since we know $\lim_{x\to a} \frac{f'(x)}{g'(x)} = L$, then
$$\lim_{x\to a} \frac{f'(c_x)}{g'(c_x)} = \lim_{x\to a} \frac{f(x)}{g(x)} = L$$

(b) Yes, the same proof works.

12. If f is twice differentiable on an open interval containing a and f'' is continuous at a, show

$$\lim_{h\to 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} = f''(a)$$

Solution: Let $F(h) = f(a+h) - 2f(a) + f(a-h)$ and $g(h) = h^2$. We have $F(0) = g(0) = 0$, both differentiable on the interval, and $g'(h) \neq 0$ for $h \neq 0$. This satisfies the conditions for L'Hopital's Rule. Applying the rule, we get

$$\lim_{h\to 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} = \lim_{h\to 0} \frac{f'(a+h) + f'(a-h)}{2h}$$

This is again 0/0. Applying the rule a second time

$$\lim_{h\to 0} \frac{f'(a+h) + f'(a-h)}{2h} = \lim_{h\to 0} \frac{f''(a+h) - f''(a-h)}{2}$$

Since f'' is continuous at a, we have

$$\lim_{h\to 0} \frac{f''(a+h) - f''(a-h)}{2} = \frac{f''(a) + f''(a)}{2} = f''(a)$$